

NightWise, explained simply

How we built it, what it currently does, and how to tell its story in a two-minute reel. Prepared 16 September 2026 from the reviewed project source, commit ec32f10, Android source version 2.3.0.

The idea in one sentence

NightWise helps people compare routes using nearby activity and road information, instead of choosing only by travel time.

Imagine leaving work late. Two roads reach the same destination, but one passes more listed shops, pharmacies and other help points. NightWise puts that information beside the route so you can make a more informed choice. It cannot promise that a road is safe.

The example we are building around

The current focus is the Bengaluru area around AEOS and Manyata Tech Park, with a roughly 20 km pilot area. A pilot is a limited first release: like trying a new restaurant menu in one branch before offering it everywhere. Kanpur is an intended expansion, but is not enabled throughout the active service yet.

What this PDF contains

Pages 2-3 explain the build and each service in plain language. Pages 4-5 list the features and explain opening times, scores and cameras. Page 6 records the limits and release checks. Pages 7-9 provide the reel priorities, a complete timed script, filming directions and a practical release plan.

Which map system is actually in this version?

Geoapify supplies map and place information. MapLibre draws the interactive map. OpenStreetMap contributes saved road and camera information. Mappls is currently an external navigation option, not the service powering the map inside this build. The earlier Google version is preserved as a recovery checkpoint.

A note on the reel goal

The aim is a video people understand, finish and want to share with a friend. Nobody can guarantee a genuinely viral result. The plan below is a creative proposal based on the product and platform guidance, not a claim that any hook or posting trick guarantees reach.

From an idea to a phone app

1. Built the journey screen

We made screens for choosing a starting point, destination, travel mode and departure time. React builds these screens from reusable pieces, like using the same door design in different rooms. TypeScript checks how pieces of code fit together and catches some mistakes before the app runs.

2. Added the map and route choices

MapLibre draws the map, roads, coloured pins and route lines. Geoapify supplies the underlying information. Map tiles are small pieces of a map image that fit together like a jigsaw puzzle. The app loads the pieces needed for the area you are viewing.

3. Put the data work behind the app

The backend is the part that works behind the screen, like a restaurant kitchen. Ours uses Node and Fastify to receive a journey, check it, ask data providers for results and send a response back. Node runs the JavaScript code on the server; Fastify organises incoming requests.

4. Connected the providers

An API is a way for one program to ask another for information. Think of NightWise asking a service desk: "Which roads connect these two points, and what places are nearby?" An API key identifies our project to that provider, like an account card. Private server keys belong behind the app, not in a public document or reel.

5. Turned listings into route information

The server checks small areas along the returned routes, removes repeated places and evaluates available opening schedules around the time the person is expected to pass. It then calculates the Night Activity Score using the evidence available for comparison.

6. Made it usable on Android and the web

Vite prepares the web code for release. Capacitor puts that web app inside an Android app and connects it to phone features such as location and the navigation-app chooser. An APK is the installable Android file, much like an installer on a computer.

7. Kept previous versions recoverable

Git tracks code changes, and GitHub stores the shared repository. A saved tag acts like a named checkpoint in a game. The Google and earlier Geoapify checkpoints let us return to an older implementation if needed. They do not automatically restore hosting settings or rebuild an APK.

One search, step by step

Choose and confirm

The person selects two places or uses phone location, checks the pins and confirms the journey. Search suggestions and address lookup help identify a place. Address lookup means converting map coordinates into a readable address; when that fails, coordinates and an entrance-check message remain useful.

Find different roads

Geoapify is asked for different route preferences. Very similar paths are merged so the screen does not pretend that almost identical roads are meaningful alternatives. The current code can return up to four distinct routes; it cannot promise three on every search.

Look beside those roads

The server groups nearby searches into small areas shared by overlapping routes. It filters results near points along the route, usually within 150 metres of a sampled point. A sample is a checkpoint on the line, like checking several stops along a bus journey. It does not search every business in the entire 20 km circle.

Check schedules and compare

Opening schedules are evaluated at the estimated passing time. The app counts places and combines available road and activity evidence into the score. Partial or missing information can reduce what the comparison can tell us.

Reuse recent results to control costs

Caching means keeping a recent answer for reuse, like remembering a price instead of calling the shop again immediately. Provider responses are kept in server memory for five minutes; map images can be reused for 24 hours. After the five-minute window, a new request fetches again. This is not automatic background monitoring.

Remember spending limits with Upstash

Upstash Redis is our persistent usage notebook. It remembers request counts even if the server restarts and helps prevent requests from exceeding the allowance we configured. It is not the five-minute shop-results store in this version. Redis is a fast data store; here it acts like a shared tally counter.

Keep the service reachable with Render

Render is the hosting service: a rented computer connected to the internet that runs the website and backend. HTTPS is the secure connection used to reach it. Public hosting can work across networks, but still depends on a healthy deployment, provider access, credits and the supported journey area.

Other supporting tools: the `opening_hours` library reads written schedules; automated checks inspect types and expected behaviour. A library is reusable code written for a specific job, like buying a ready-made clock mechanism rather than building one from scratch.

What a person can use

Plan a journey

Choose origin and destination; use current location; swap endpoints; use the AEOS-Manyata preset; review pins and entrances; choose car, bike or walking; choose extra travel time to consider. Car and leave-now are the defaults. The screen allows departure up to five hours ahead, with the limitation explained on page 6.

Compare route options

See travel time, distance and extra time; select route cards; tap a route line on the map to choose it; view activity details. Near-duplicate routes are collapsed. Current labels include Fastest, Safest, Fastest and safest, and Alternative; the Safest wording has an evidence-consistency issue noted on page 6.

Explore the map

Zoom and move around; expand to full screen and minimise again; see start and destination pins and a colour legend. Shops are yellow, pharmacies/clinics pink, hospitals purple, petrol/CNG green, and mapped cameras cyan. The route also shows highlighted low-activity sections when available.

Inspect places and opening information

View found totals, open or estimated-open counts where available, opening details, help points, opening-times notes and data-source notes. The default place display focuses on open or estimated-open places. Missing hours do not make a place verified open; unresolved places remain part of the found total.

See the activity score

Route cards use a circular score out of 100, with the existing ring presentation and activity details. The score depends on available evidence; some results are estimates. It is our calculation, not a score supplied by Google, Geoapify or a police authority.

See additional mapped places and cameras

The build includes 205 supplementary saved place records and 2,803 saved camera records for the pilot area. Only nearby matching records are displayed for the selected route. These are dataset totals, not counts that will appear on every journey.

Refresh, recover and continue

Refresh live results, see when the comparison was checked, retry failed requests, use tutorial scenarios, check the service connection, provide simple usefulness feedback and continue to a navigation app. There is no compulsory startup team-code screen; a server access-code setting can still require a code in Settings.

Android opens a destination-app chooser for compatible installed apps. The browser uses a Mappls directions link. Neither should be described as guaranteed support for every navigation app or exact preservation of every selected route.

Hours, scores and cameras

Opening hours: a schedule, not a live shutter sensor

If a provider supplies usable hours, those take priority. If it says 24/7, the app can show scheduled always-open availability. This does not prove that staff are there right now. Special holidays and unsupported schedules can remain unresolved.

When usable hours are missing, the current planning defaults are shops/food 9 am-8 pm, pharmacies 10 am-11 pm, and fuel stations 24 hours, in Indian time. These are estimates. Hospitals do not receive a blanket 24-hour default. A fresh API response does not make an old business listing newly verified.

The six ingredients of the score

Open place density: 25 points. How many open places are found around the route. Density simply means how closely places are spread together.

Main road share: 20 points. How much of the route uses roads classified as main roads.

Help points: 15 points. Available evidence about places that may help a traveller.

Activity continuity: 15 points. Whether activity continues along the journey or has long gaps. Think of street lights spread along a road rather than all clustered at one end.

Route simplicity: 15 points. How straightforward the path is. Open transport points: 10 points. Available evidence about open transport-related places.

The comparison uses evidence available across the compared routes and scales the result to 100. If hours are sparse, the estimated calculation drops some components. A score of 70 is not a 70% chance of safety; it is a relative activity-and-road comparison.

What the camera layer really shows

The saved OpenStreetMap extract was taken from source data timestamped 15 September 2026, 23:03:27 UTC, and retrieved on 16 September. The app shows matching records within 60 metres of a selected route. This is not live CCTV video or a verified police traffic-camera feed. Some records may be old, private or no longer operational.

Cameras currently do not add points to the six-part numerical score. Camera counts can influence which route gets the Safest label when scores are within three points. This needs a consistent rule before being promoted as a safety feature. More mapped cameras do not establish lower crime risk.

What is still unfinished

Navigation does not fully carry the chosen journey

The Android chooser sends destination coordinates and a name. It does not send the selected starting point, travel mode or route stops. The receiving app may calculate a different route. The web link can carry route points, but does not carry travel mode in this implementation.

Five hours ahead is not future traffic prediction

The screen accepts a later departure, but the routing request does not send that chosen time to Geoapify. It uses approximated traffic. Future opening-time estimates and future traffic predictions are different things.

Kanpur and worldwide journey coverage are not ready

The active validation, search, map limits and road evidence remain centred on Bengaluru. A phone being able to reach the server from another network is different from supporting routes everywhere.

The Safest label needs correction

Estimated comparisons can still receive this label. Camera tie-breaks can disagree with numerical ranking, and the pairwise sorting rule can produce inconsistent ordering. The previous analysis documented this; it did not fix the code.

Data needs field checks and refreshes

Saved camera data needs to be regenerated to update it. Shops and entrances need spot checks on the ground. We should verify examples before recording the reel, especially any claimed open business or camera location. Do not imply that all 2,803 records were checked today.

Verification is not yet fully clean

In the preceding source review, TypeScript passed and 231 of 232 automated tests passed. One address-display test failed because it expected a different provider path. No new phone test, deployment acceptance check or APK build was performed for this PDF. The source commit is unchanged from that review.

Before sharing an installation link

Confirm the deployed version, service health, API allowance and access-code configuration. Test on a real phone with mobile data, confirm the navigation behaviour and resolve the failing test. Keep private keys out of screenshots. Preserve the Google recovery bookmark. Update the older README and hosting instructions so they match the active provider.

What the reel should promise

Use: "Compare nearby activity before you choose a route." Avoid: "Guaranteed safest route", "live CCTV coverage", "all shops verified open", "works everywhere" or "keeps the exact route in any navigation app". The current build does not establish those claims.

Make the benefit visible

The central idea

"At night, would you take the shortest road, or first check what is along it?" This is a relatable decision, not a technical lecture. Lead with a real route screen, then show the extra information NightWise adds. Keep the audience focused on one journey: AEOS to Manyata.

Must mention or show, in priority order

1. The late-night travel question. 2. One real route comparison with actual minutes. 3. Shops, medical help and fuel pins. 4. Scheduled versus estimated opening status. 5. The circular Night Activity Score, described as a comparison. 6. Mapped cameras, clearly labelled as saved locations. 7. Full-screen route selection. 8. Bengaluru pilot status and one clear invitation for feedback.

The builder story: one sentence is enough

"We connected maps, place schedules and our own comparison logic, then packaged it into an Android app." Explain the providers in the PDF or a follow-up video. Listing React, Fastify, Redis and every API in the main reel uses time without showing the viewer why they should care.

Three opening lines to try

A: "At 11 pm, would you choose a road only because it is two minutes faster?" Use two minutes only as a question, not as a fabricated result on the screen. B: "Before I choose a route at night, I want to see what is along it." C: "I built a route comparison app around one question: what will I pass on the way?"

Why this could be shared

A late-shift worker can recognise the problem immediately and send the video to a colleague. The moving map provides visible proof of the feature. The building story gives it a human owner. Honest limits make the demonstration easier to trust. These are creative reasons to try the concept, not proven guarantees of performance.

What platform guidance supports

Meta describes Instagram recommendation systems as personalised rather than one fixed algorithm. Its published Instagram guidance favours original versions when duplicate content is found. Use your own screen recording and narration. The timing, hooks and shot choices here are editorial recommendations, not official ranking rules. Sources are linked on page 9.

A complete 120-second story

About 245 spoken words. Rehearse at a comfortable pace; the time slots include breathing room and screen actions. Do not rush the explanation to match the clock.

0-10 seconds | Hook

At eleven at night, would you choose a road only because it is a little faster? I wanted to see what was along the journey, too.

Visual: Show a real route comparison immediately. Add: "What is along your route?"

10-25 seconds | Introduce the app

So we built NightWise. Let me show you AEOS to Manyata Tech Park. Choose your starting point and destination, confirm the pins, and compare the available routes.

Visual: Record the two endpoints, confirmation and returned options. Keep actual times visible.

25-43 seconds | Show places

Tap a road and the map changes with it. You can see listed shops, pharmacies, hospitals and fuel stops near that route, then open the map full screen for a closer look.

Visual: Tap an alternative, expand the map and briefly open a real place pin.

43-59 seconds | Explain hours

We check available opening schedules around the time you are expected to pass. When hours are missing, some categories use labelled estimates. A listing cannot confirm that a shop is physically open.

Visual: Show a scheduled example and an estimated example. Never replace missing hours with invented hours.

59-76 seconds | Explain the score

This circle is our Night Activity Score. It compares available place activity and road information. It is not a guarantee of safety, and missing information affects what we can compare.

Visual: Show the score ring and activity details. Overlay: "Activity estimate, not a safety guarantee."

76-91 seconds | Camera reveal

We also show mapped camera locations near the chosen route. These are saved map records, not live CCTV feeds, and they do not confirm that a camera is working.

Visual: Zoom to the cyan camera pins. Overlay: "Mapped locations - not live video."

91-108 seconds | Builder payoff

Behind the screen, we connected map services, place data and our own comparison logic. Recent answers are reused for five minutes to reduce repeat requests. Then we packaged the experience for Android.

Visual: Use three simple labels: "Routes / Places / Comparison"; finish on the app running on a phone.

108-120 seconds | Close

We are starting around AEOS and Manyata in Bengaluru. If you travel after dark, what information would help you choose? Tell us what you would want to see along your route.

Visual: Show the full route and NightWise name. Leave the question readable through the final frame.

Give the video its best chance

Recording checklist

Record vertically with readable app text. Use a real journey response rather than a mock result. Crop or blur personal location details and secret keys. Capture the app screen separately from the voice so you can keep both clear. Avoid a long logo animation before the demonstration.

Use captions for the spoken explanation, one short thought at a time. Change framing when the subject changes: route, place, score, camera. Keep map attribution visible where required. If you use music, choose audio you have permission to use and keep it below the voice. Do not add fake warning sirens or staged danger to sell the app.

Cover and caption

Cover: "What is along your night route?" Use one clear map crop with a few readable pins. Suggested caption: "We are building NightWise to compare nearby activity along a journey, starting around AEOS and Manyata in Bengaluru. Opening times may be estimates; camera pins are saved mapped locations. What would help you choose a route after dark?"

What to leave for another video

The full backend stack, precise score weights, Upstash settings, installation instructions and provider migration history belong in separate posts. Do not cram all features into two minutes. The PDF is the complete explanation; the reel is one clear reason to care.

Learn from the release

Review average watch time, completion where available, shares, saves and useful comments after the video has had time to circulate. Compare them with your own similar videos rather than a made-up universal target. If viewers leave before the map demo, shorten the setup. If they mistake camera pins for live feeds, make the explanation clearer. Test a different opening in a later edit; do not assume repeated uploads guarantee distribution.

A useful follow-up is a real, carefully documented field check of one route: do the entrances and listed places match? Another is a short explanation of why the score is an estimate. Let actual audience questions choose the next video rather than inventing engagement.

Sources and scope

Project evidence: the source review saved in `talks/2026-09-16_project-analysis_geoapify-restore_v01.md`, based on commit `ec32f10`. Relevant source includes the Geoapify server, comparison and route-label logic, camera data, map renderer and Android navigation bridge. This PDF explains that review; it is not a new live-data certification.

Meta: [Helping creators find new audiences \(Instagram, 2024\)](#)

Meta: [How AI influences Facebook and Instagram \(2023\)](#)

Platform sources checked 16 September 2026. They support the originality and personalised-recommendation points on page 7. The creative script and release plan are our proposed approach; no reach, view or follower outcome is promised.